

Gi-Luen (Allen) Huang

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SUMMARY

Senior AI Engineer & Assistant Project Manager with a strong research foundation in computer vision and deep learning (MS, NTU), now with 3 years of industry experience spanning generative AI and automotive AI. At CyberLink, I independently developed generative AI features with Flux-based models—training LoRA and ControlNet for object removal & replacement, product background generation, and human beauty effects (Flux Kontext). I am now guiding team members in shipping generative AI features and driving the team's adoption of agentic development workflows. Previously, at MobileDrive, I developed and deployed automotive AI on the Qualcomm 8295 platform for on-vehicle testing.

EDUCATION

National Taiwan University

Feb. 2021 - Jan. 2023

MS in Data Science Group of Communication Engineering, GPA: 4.30/4.30

Taipei, Taiwan

Transcript

- **Courses:** Machine Learning, Deep Learning for Computer Vision, Applied Deep Learning, Deep Learning for Human Language Processing, Computer Vision, Convex Optimization, Time-Frequency Analysis and Wavelet Transform
- **Thesis:** "CTGAN: Cloud Transformer Generative Adversarial Network"
- **Advisor:** *Prof. Pei-Yuan Wu*

National Taiwan University of Science and Technology

Jun. 2017 - Jan. 2021

BS in Electrical Engineering, GPA: 4.09/4.30

Taipei, Taiwan

Transcript

- **Courses:** Data Structures, Algorithm design and analysis, Programming
- **Paper publication:** "Face Expression and Tone of Voice for Deception System"
- **Advisor:** *Prof. Jing-Ming Guo*

WORK EXPERIENCES

CyberLink

New Taipei, Taiwan

Senior AI Engineer & Assistant Project Manager

May 2026 - Present

- Guide team members in developing GenAI-powered features, and assist in planning feature roadmaps and release timelines.
- Established and shared best practices for coding-agent adoption, helping the team reduce unexpected outputs and improve development efficiency.
- Designed and built an internal Agentic AI PoC platform integrated with the company's authentication system, enabling PMs and designers to accelerate their product and design workflows.

Senior AI Engineer

May 2025 - May 2026

- Trained LoRA on Flux for adaptive product background generation and object removal.
- Trained LoRA on Flux Kontext to deliver a human beauty-effect feature, whose quality outperformed Meitu in internal QA benchmarks.
- Built an image-to-video feature based on Wan2.2-Animate.

AI Engineer

Aug. 2024 - May 2025

- Developed Flux-based features trained with LoRA and ControlNet, including text-to-image generation and product background generation.
- Built and shipped the generative fill feature, which became the most-used feature in PhotoDirector Mobile (PhDM) after launch; in internal QA benchmarks, its quality outperformed Adobe Firefly, SnapEdit, and TouchRetouch.

MobileDrive Technology

Machine Learning Engineer

Jun. 2023 - May 2024

New Taipei, Taiwan

- Developed a method for projecting future vehicle trajectories on current front-view images as ground-truth for 3D Vehicle Future Trajectory Model. This approach utilizes data collected by vehicles equipped with RTK and front-view cameras, enabling model training **with minimal human intervention**.
- Developed and optimized 3D Vehicle Future Trajectory Model, significantly enhancing performance through empirical experiments focused on dataset improvement, warping method, and data augmentation. Conducted model quantization for deployment on embedded devices.
- Deployed the quantized model on the **Qualcomm 8295 platform** by converting the Python program to C++. Optimized model efficiency achieved **200 FPS**, and the quantization error (L1 error) was reduced from 4.0 to 0.15 (**266% ↓**).
- Collaborated with team members to integrate the quantized model into the AR navigation algorithm, enabling **on-vehicle testing**. This enhancement significantly improved performance in challenging scenarios, such as navigating roundabouts and executing sharp turns.
- Developed intersection depth estimation **auto-labeling algorithm** using projected RTK trajectory.

Jubo Health

Machine Learning Engineer Intern

Jul. 2022 - Aug. 2022

New Taipei, Taiwan

- Implemented general frameworks to support various deep learning tasks including recognition, segmentation, and object detection.
- Enhanced the existing Wound Classification Model, improving accuracy by **approximately 3%**.
- Deployed models as a service using Docker on GCP.

Neurobit Technologies

Machine Learning Engineer Intern

Feb. 2022 - Jun. 2022

Taipei, Taiwan

- Developed Gaze Estimation Model using a self-supervised learning technique, successfully **reducing the error rate by 90%**.
- Developed a feature matching algorithm to accurately detect torsional eye rotation.

Taiwan Semiconductor Manufacturing Company (TSMC)

Information Technology (IT) Intern

Jul. 2021 - Aug. 2021

Hsinchu, Taiwan

- Full-stack system integration
- Implemented backend functionality for a webpage that changes color in response to user button clicks.

SKILLS

Programming	Python (main), C++
Generative AI	Diffusion models (Flux, Flux Kontext), LoRA, ControlNet, image-to-video (Wan2.2)
Deep Learning	PyTorch, model quantization, edge deployment (Qualcomm 8295)
Tools & Agentic	Git, Docker, AI coding agents (Claude Code, Codex)

TA EXPERIENCES

NTU - Deep Learning for Computer Vision

MS student in Graduate Institute of Communication Engineering

Advisor: Prof. Yu-Chiang Frank Wang

2022 Fall
Taipei, Taiwan

- Designed and graded homework sets
 - [Generative Adversarial Network \(GAN\)](#)
 - [Conditional Diffusion models \(DDPM\)](#)
 - [Domain Adaptation model \(DANN\)](#)
 - [Final project: 3D Indoor Scene Long Tail Segmentation](#)
- Motivated students during TA office hours

ITRI - Machine Learning

MS student in Graduate Institute of Communication Engineering

Advisor: Prof. Pei-Yuan Wu

Sep. 2022 - Oct. 2022
Hsinchu, Taiwan

- Designed programming exercises
 - [PM2.5 prediction \(Regression model\)](#)
 - [Income prediction \(Classification model\)](#)
 - [Facial Emotion Recognition](#)
 - [Text Sentiment Classification](#)
 - [Dimension Reduction](#)
 - [Image Event Anomaly Detection](#)

NTU - Time-Frequency Analysis and Wavelet Transform

MS student in Graduate Institute of Communication Engineering

Advisor: Prof. Jian-Jiun Ding

2021 Fall
Taipei, Taiwan

- Graded the homework sets

NTU - Data Structure

MS student in Graduate Institute of Communication Engineering

Advisor: Prof. Pei-Yuan Wu

2021 Spring
Taipei, Taiwan

- Designed and graded the theoretical homework set
 - [Big-O notation definition](#)
 - [Red-black tree](#)
 - [Disjoint sets](#)
 - [AA tree](#)
- Designed and graded the programming homework set
 - [Dynamic Programming \(DP\)](#)
 - [Tree data structure implementation](#)

AI COMPETITIONS

[Orchid Species Identification and Classification](#)

2022 T-Brain Competition

Apr. 2022 - Jun. 2022
[Github Link](#)

- Applied ConvNext and Swin_transformer to conduct image recognition task
- Applied data augmentation methods to enhance models' generalization ability, including random crop, random rotation, Mixup, random erasing, etc.
- **Private Leaderboard: 14/743, Top 3%**

[Lung Adenocarcinoma Pathological image segmentation](#)

2022 T-Brain Competition

Mar. 2022 - Jun. 2022
[Github Link](#)

- Developed Deeplab-v3-plus to segment the cells having STAS features
- Developed the post-processing method to fill in holes after model prediction
- Applied data augmentation methods to enhance the models' robustness, including horizontal/vertical flip, random rotation, color jitter, etc.
- **Private Leaderboard: 2/307, Top 1%**

Crops Status Monitoring by Image Recognition

Mar. 2022 - May 2022

2022 AIda Competition

[Github Link](#)

- Developed ConvNext and Resnet50 models to do ensemble prediction
- Applied data augmentation methods during training, including horizontal/vertical flip, affine transformation, etc.
- Applied Grad-cam to visualize the attention location of model prediction
- **Private Leaderboard: 3/428, Top 1%**

Human Voice Denoising

Feb. 2022 - May 2022

2022 AIda Competition

[Github Link](#)

- Based on U-net, developed a 1d-convolutional neural network as an autoencoder
- Applied data augmentation methods during training, including reverb, remix, shift, etc.
- Combined time domain and frequency domain loss functions
- **Private Leaderboard: 6/282, Top 2%**

Traditional Chinese Scene Text Recognition (Advanced)

Nov. 2021 - Dec. 2021

2021 T-Brain Competition

[Github Link](#)

- Applied Yolov5 for signboard detection
- Developed Resnet18 model to conduct ROI transformation
- Developed modified Vision Transformer to conduct text recognition
- Applied data augmentation methods during training, including horizontal/vertical flip, affine transformation, resolution transformation, etc.
- **Private Leaderboard: 6/128, Top 5%**

Traditional Chinese Scene Text Recognition (Intermediate)

Aug. 2021 - Oct. 2021

2021 T-Brain Competition

[Github Link](#)

- Applied Yolov5 to capture the subword from a word
- Developed arcMargin loss function on Resnet18 model
- Applied data augmentation methods during training, including horizontal/vertical flip, affine transformation, resolution transformation, etc.
- **Private Leaderboard: 5/183, Top 3% and Innovation Award from T_brain**

Rice Plant Position Labeling in UAV full-color image

Aug. 2021 - Oct. 2021

2021 AIda Competition

- Applied Yolov5 to capture the rice plant location
- Applied rule-based post-processing to deal with the overlapping region after the sliding window method
- Applied data augmentation methods during training, including horizontal/vertical flip, affine transformation, resolution transformation, etc.
- **Private Leaderboard: 18/523, Top 3%**

SELECTED PROJECTS

Pupil Tracking

NTU - Computer Vision (Final Project)

Instructor: Prof. Shao-Yi Chien

2022 Spring

[Github Link](#)

- Combined the deep learning model Deeplab-v3-plus with the traditional CV method to obtain pupil segmentation.
- **Private leaderboard: 3/21, Top3**

Intracranial Hemorrhage Prediction

NTU - Application of Deep Learning in Medical Imaging

Instructor: Prof. Joe Yeh

2021 Fall

[Github Link](#)

- Developed an ensemble model of Resnet50 and SEresnet50 to conduct multi-label classification problem

Adversarial Attack on Deception Detection

NTU - Security and Privacy of Machine Learning (Final Project)

Instructor: Prof. Shang-Tse Chen

2021 Fall

[Github Link](#)

- Designed experiments about the adversarial attack on deception detection

Fine-grained Food Classification

NTU - Deep Learning for Computer Vision (Final Project)

Instructor: Prof. Yu-Chiang Frank Wang

2021 Fall

[Github Link](#)

- Designed a cosine sampling method to deal with the data imbalanced problem
- Implemented ArcFace loss function so that the extracted features can be separated in high-dimensional space

PUBLICATIONS

Chen, P. W., Yang, T. S., Huang, **G. L.**, **Huang** *et al.*(2023). Viewing Bias Matters in 360° Videos Visual Saliency Prediction. IEEE Access.

Huang, Gi-Luen and Pei-Yuan Wu (2022). "CTGAN : Cloud Transformer Generative Adversarial Network". In: *2022 IEEE International Conference on Image Processing (ICIP)*, pp. 511–515. DOI: [10.1109/ICIP46576.2022.9897229](https://doi.org/10.1109/ICIP46576.2022.9897229).

Li-Wei Hsiao, Jing-Ming Guo, **Gi-Luen Huang**, *et al.* "Face Expression and Tone of Voice for Deception System". 2020 International Conference on System Science and Engineering (ICSSE), 2020 (Best student paper award)

AWARD/SCHOLARSHIPS

Pan Wen Yuan Foundation Scholarship

MS student in Graduate Institute of Communication Engineering

2022

Taipei, Taiwan

Outstanding Student

Department of Electrical Engineering at NTUST

2021

Taipei, Taiwan